

Math Thermo
class # 26
4/30/2026

Binary Quasi-Compressible Flow

Recall the model from Lectures 23 and 24. Keep all of the assumptions the same, except let us assume that the partial densities satisfy

$$\begin{aligned} \text{where} \quad \rho_i &= \rho_i^0 \phi_i, \quad i=1,2, \\ \text{and} \quad \rho_i^0 &> 0 \quad (\text{distinct constants}), \quad i=1,2, \\ &\phi_1 + \phi_2 = 1. \end{aligned}$$

The total mass density is

$$\rho := \sum_{i=1}^2 \rho_i = \sum_{i=1}^2 \rho_i^0 \phi_i.$$

Assume that the mass equations satisfy, $i=1,2$,

$$\rho_i^0 \frac{\partial \phi_i}{\partial t} + \rho_i^0 \nabla \cdot (\phi_i \vec{v}) = -\nabla \cdot \vec{J}_i^D; \quad \sum_{i=1}^2 \vec{J}_i^D = \vec{0}.$$

Summing this equation, we have, as desired,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0.$$

Now, dividing by $\rho_i^0 > 0$, we also have

$$\frac{\partial \phi_i}{\partial t} + \nabla \cdot (\phi_i \vec{v}) = \frac{-\nabla \cdot \vec{J}_i^D}{\rho_i^0}.$$

$$\text{Summing,} \quad \nabla \cdot \vec{v} = \frac{-\nabla \cdot \vec{J}_1^D}{\rho_1^0} - \frac{-\nabla \cdot \vec{J}_2^D}{\rho_2^0} = \left(\frac{\rho_2^0 - \rho_1^0}{\rho_1^0 \rho_2^0} \right) \nabla \cdot \vec{J}_2^D.$$

Repeat the calculations of lectures 23 and 24 with the change above. What are the model equations?

Unary Compressible Flow

Let us consider a unary ($r=1$), non-reacting (obviously) flow. Then,

$$\sigma_S = \sigma_{S,0} + \sigma_{S,1} + \sigma_{S,2}$$

where

$$\sigma_{S,0} := - \frac{p \nabla \cdot \vec{v}}{3T},$$

$$\sigma_{S,1} := \vec{J}_v^D \cdot \nabla \left(\frac{1}{T} \right),$$

$$\sigma_{S,2} := - \frac{\tau : \dot{\Pi}}{T}.$$

We are guaranteed non-negative entropy production if

$$p = -\kappa \nabla \cdot \vec{v}, \quad \kappa > 0,$$

$$\vec{J}_v^D = M T^2 \nabla \left(\frac{1}{T} \right), \quad M > 0,$$

$$\dot{\Pi} = -\mu_0 \tau, \quad \mu_0 > 0.$$

The evolution equations are

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) &= 0, \\ \frac{\partial (\rho \vec{v})}{\partial t} + \nabla \cdot (\rho \vec{v} \otimes \vec{v}) + \nabla p &= \nabla \cdot (\mu_0 \tau + \kappa \nabla \cdot \vec{v}) \\ &\quad - \sum_{i=1}^r \rho_i \nabla \psi_i, \end{aligned}$$

where

$$\tau := \frac{1}{2} (\nabla \vec{v} + (\nabla \vec{v})^T) - \frac{1}{3} \nabla \cdot \vec{v} \mathbb{I}_3,$$

$$\frac{\partial}{\partial t}(\rho u) + \nabla \cdot (\rho u \vec{v}) = M \Delta T - P \nabla \cdot \vec{v} + \nabla \vec{v} : (\mu_0 \mathbb{I} + \kappa \nabla \cdot \vec{v}).$$

It is customary to have

$$\hat{u} = \rho u = \rho C T, \quad C > 0.$$

Then,

$$\frac{\partial}{\partial t}(\rho u) + \nabla \cdot (\rho u \vec{v}) = C \rho \frac{DT}{Dt},$$

so that

$$C \rho \frac{DT}{Dt} = M \Delta T - P \nabla \cdot \vec{v} + \nabla \vec{v} : (\mu_0 \mathbb{I} + \kappa \nabla \cdot \vec{v})$$

is our heat transport equation.

To close the system, we need boundary conditions and we need an equation of state relating the hydrostatic pressure with the density and temperature. For example,

$$P = \gamma_p T, \quad \gamma > 0,$$

or

$$P = \gamma_p \rho + \gamma_T T, \quad \gamma_p, \gamma_T > 0.$$

Thermo-Diffusion of Binary Mixtures

In this setting, we assume that the flow is diffusion dominated, i.e.,

$$\vec{v} \approx \vec{0}.$$

The mixture, we assume, is comprised of two components, $r=2$. And, for simplicity, we assume that potential energies can be neglected and that there are no reactions. Let us also assume that the total mass density is constant, i.e.,

$$\rho(\vec{x}, t) = \rho_0 > 0, \quad \forall \vec{x} \in \Omega, t \geq 0.$$

The evolution equations are

$$\begin{aligned} \phi_1 + \phi_2 &= 1, \\ \rho_0 \frac{\partial \phi_i}{\partial t} &= -\nabla \cdot \vec{J}_i^D, \quad (i=1, 2) \\ \frac{\partial \hat{u}}{\partial t} &= -\nabla \cdot \vec{J}_U^D. \end{aligned}$$

The partial densities are given by

$$\rho_i = \phi_i \rho_0$$

so that

$$\rho = \rho_1 + \rho_2 = \rho_0(\phi_1 + \phi_2) = \rho_0,$$

as desired. If we sum the mass equations,

we get

$$0 = \frac{\partial \rho_0}{\partial t} = \rho_0 \frac{\partial \phi_1}{\partial t} + \rho_0 \frac{\partial \phi_2}{\partial t} = -\nabla \cdot (\vec{J}_1 + \vec{J}_2).$$

We will continue to assume that

$$\vec{J}_1^D + \vec{J}_2^D = \vec{0},$$

which is consistent with the last equation.

The entropy equation becomes

$$\frac{\partial \hat{S}}{\partial t} = -\nabla \cdot \vec{J}_S^D + \sigma_S,$$

where

$$\begin{aligned}\vec{J}_S^D &= \frac{1}{T} \left(\vec{J}_U^D - \sum_{i=1}^2 \mu_i \vec{J}_i^D \right), \\ \sigma_S &= \vec{J}_U^D \cdot \nabla \left(\frac{1}{T} \right) - \sum_{i=1}^2 \vec{J}_i^D \cdot \nabla \left(\frac{\mu_i}{T} \right).\end{aligned}$$

Defining

$$\tilde{\mu}_2 = \mu_2 - \mu_1$$

and using $\vec{J}_1^D + \vec{J}_2^D = \vec{0}$, we can write

$$\begin{aligned}\vec{J}_S^D &= \frac{1}{T} \left(\vec{J}_U^D - \tilde{\mu}_2 \vec{J}_2^D \right), \\ \sigma_S &= \vec{J}_U^D \cdot \nabla \left(\frac{1}{T} \right) - \vec{J}_2^D \cdot \nabla \left(\frac{\tilde{\mu}_2}{T} \right).\end{aligned}$$

Using the identity

$$\nabla \left(\frac{\tilde{\mu}_2}{T} \right) = \frac{1}{T} \nabla \tilde{\mu}_2 + \tilde{\mu}_2 \nabla \left(\frac{1}{T} \right),$$

we find

$$\begin{aligned}\sigma_s^D &= \vec{J}_v^D \cdot \nabla\left(\frac{1}{T}\right) \\ &\quad - \vec{J}_2^D \cdot \left(\frac{1}{T} \nabla \tilde{\mu}_2 + \tilde{\mu}_2 \nabla\left(\frac{1}{T}\right)\right) \\ &= \nabla\left(\frac{1}{T}\right) \cdot (\vec{J}_v^D - \tilde{\mu}_2 \vec{J}_2^D) \\ &\quad - \frac{1}{T} \vec{J}_2^D \cdot \nabla \tilde{\mu}_2 \\ &= T \vec{J}_s^D \cdot \nabla\left(\frac{1}{T}\right) - \frac{1}{T} \vec{J}_2^D \cdot \nabla \tilde{\mu}_2.\end{aligned}$$

Using the Onsager reciprocal relations, we have

$$(26.1) \quad \vec{J}_s^D = L_{1,1} T \nabla\left(\frac{1}{T}\right) - L_{1,2} \frac{1}{T} \nabla \tilde{\mu}_2,$$

$$(26.2) \quad \vec{J}_2^D = L_{2,1} T \nabla\left(\frac{1}{T}\right) - L_{2,2} \frac{1}{T} \nabla \tilde{\mu}_2,$$

where

$$L_{2,1} = L_{1,2}.$$

To guarantee that $\sigma_s \geq 0$, we require that

$$L = \begin{bmatrix} L_{1,1} & L_{1,2} \\ L_{2,1} & L_{2,2} \end{bmatrix}$$

is SPD.

Notice carefully what (26.1) is telling us. If $L_{1,2} \neq 0$, a gradient in the chemical potential $\tilde{\mu}_2$ can lead to a flow of heat energy, even if the temperature is constant (and therefore has no gradient).

What does (26.2) tell us? If $L_{1,2} \neq 0$, a gradient in the temperature field leads to a flow of mass, even if the chemical potential is constant (and therefore has no gradient).

Are these scenarios physical?

In the simplest diagonal setting $L_{1,2} = L_{2,1} = 0$ and $L_{11} > 0$, $L_{22} > 0$. Then

$$\vec{J}_s^D = L_{1,1} T \nabla \left(\frac{1}{T} \right),$$

$$\vec{J}_2^D = -L_{2,2} \frac{1}{T} \nabla \tilde{\mu}_2,$$

and again $\sigma_s \geq 0$ is guaranteed. This setting is for many cases too simplistic and too restrictive.

Thermal Diffusion

Consider a unary material (no reactions) wherein heat transport is diffusion dominated, i.e.,

$$\vec{v} \approx \vec{0}.$$

Let us also assume that the total mass density is constant, i.e.,

$$\rho(\vec{x}, t) = \rho_0 > 0, \quad \forall \vec{x} \in \Omega, \quad t \geq 0.$$

The evolution equations are just

$$\rho \equiv \rho_0,$$

$$\frac{\partial \hat{u}}{\partial t} = -\nabla \cdot \vec{J}_u^D.$$

We, as usual take

$$\rho u = \hat{u} = \rho C T = \rho_0 C T, \quad C > 0.$$

The entropy equation is

$$\frac{\partial \hat{S}}{\partial t} = -\nabla \cdot \vec{J}_s^D + \sigma_s,$$

where

$$\vec{J}_s^D = \frac{1}{T} \vec{J}_u^D$$

and

$$\sigma_s = \vec{J}_u^D \cdot \nabla \left(\frac{1}{T} \right).$$

We take

$$\vec{J}_u^D = \kappa T^2 \nabla \left(\frac{1}{T} \right), \quad \kappa > 0,$$

which guarantees that

$$\sigma_s = \kappa T^2 \left| \nabla \left(\frac{1}{T} \right) \right|^2 \geq 0.$$

Of course, we will have

$$\rho_0 C \frac{dT}{dt} = -\nabla \cdot \left(\kappa T^2 \nabla \left(\frac{1}{T} \right) \right) = \kappa \Delta T.$$

Thus,

$$\frac{dT}{dt} = D \Delta T,$$

where

$$D = \frac{\kappa}{\rho_0 C}.$$

The heat equation is "the" heat equation.

The Cahn-Hilliard Equation

Take the same assumptions as in the last example.

Assume further that

$$\rho u =: \hat{u}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) = \tilde{u}(T) + \lambda_u \delta(\phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2),$$

$$\rho f =: \hat{f}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) = \tilde{f}(T) + \lambda_f \delta(\phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2),$$

$$\rho s =: \hat{s}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) = \tilde{s}(T) + \lambda_s \delta(\phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2),$$

with

$$\tilde{f} = \tilde{u} - T \tilde{s} \quad \text{and} \quad \lambda_f = \lambda_u - T \lambda_s.$$

We assume that, in particular,

$$\lambda_f = \lambda_u - T_0 \lambda_s > 0.$$

Recall

$$\delta = \frac{1}{\varepsilon} \frac{\phi_1^2 \phi_2^2}{4} + \frac{\varepsilon}{2} \left(\frac{|\nabla \phi_1|^2}{2} + \frac{|\nabla \phi_2|^2}{2} \right)$$

where $\varepsilon > 0$ has units of length. Setting

$$W(\phi_1, \phi_2) = \frac{\phi_1^2 \phi_2^2}{4},$$

we have

$$\delta = \frac{1}{\varepsilon} W(\phi_1, \phi_2) + \frac{\varepsilon}{2} \left(\frac{|\nabla \phi_1|^2}{2} + \frac{|\nabla \phi_2|^2}{2} \right).$$

We can assume that thermal compatibility holds:

$$\frac{\partial \hat{\delta}}{\partial T} = \frac{1}{T} \frac{\partial \hat{u}}{\partial T}.$$

But this will not matter in the present case, since

$$T = T_0 > 0$$

where T_0 is a constant. Recall that, in the isothermal setting we want the free energy to be minimized.

Consider the following:

$$\begin{aligned} \frac{\partial \hat{F}}{\partial t} &= \frac{\partial \hat{F}}{\partial T} \frac{\partial T}{\partial t} + \frac{\partial \hat{F}}{\partial \phi_1} \frac{\partial \phi_1}{\partial t} + \frac{\partial \hat{F}}{\partial \phi_2} \frac{\partial \phi_2}{\partial t} \\ &\quad + \sum_{i=1}^d \frac{\partial \hat{F}}{\partial \vec{p}_{i1}} \frac{\partial \vec{p}_{i1}}{\partial t} + \sum_{i=1}^d \frac{\partial \hat{F}}{\partial \vec{p}_{i2}} \frac{\partial \vec{p}_{i2}}{\partial t} \end{aligned}$$

where

$$\vec{p}_i = \nabla \phi_i, \quad i=1,2.$$

Thus,

$$\begin{aligned} \frac{\partial \hat{F}}{\partial t} &= \lambda_f \frac{1}{\varepsilon} \frac{\partial W}{\partial \phi_1} \frac{\partial \phi_1}{\partial t} + \lambda_f \frac{1}{\varepsilon} \frac{\partial W}{\partial \phi_2} \frac{\partial \phi_2}{\partial t} \\ &\quad + \lambda_f \frac{\varepsilon}{2} \nabla \phi_1 \cdot \nabla \left(\frac{\partial \phi_1}{\partial t} \right) + \lambda_f \frac{\varepsilon}{2} \nabla \phi_2 \cdot \nabla \left(\frac{\partial \phi_2}{\partial t} \right) \end{aligned}$$

$$\begin{aligned}
&= \lambda_f \left(\frac{1}{\varepsilon} \frac{\partial \omega}{\partial \phi_1} - \frac{\varepsilon}{2} \Delta \phi_1 \right) \frac{\partial \phi_1}{\partial t} \\
&\quad + \lambda_f \left(\frac{1}{\varepsilon} \frac{\partial \omega}{\partial \phi_2} - \frac{\varepsilon}{2} \Delta \phi_2 \right) \frac{\partial \phi_2}{\partial t} \\
&\quad + \lambda_f \frac{\varepsilon}{2} \nabla \cdot \left(\nabla \phi_1 \frac{\partial \phi_1}{\partial t} \right) + \lambda_f \frac{\varepsilon}{2} \nabla \cdot \left(\nabla \phi_2 \frac{\partial \phi_2}{\partial t} \right) \\
&= -\frac{\lambda_f}{\rho_0} \omega_1 \nabla \cdot \vec{J}_1^D - \frac{\lambda_f}{\rho_0} \omega_2 \nabla \cdot \vec{J}_2^D \\
&\quad + \lambda_f \frac{\varepsilon}{2} \nabla \cdot \left(\nabla \phi_1 \frac{\partial \phi_1}{\partial t} + \nabla \phi_2 \frac{\partial \phi_2}{\partial t} \right).
\end{aligned}$$

Using $\vec{J}_1^D = -\vec{J}_2^D, \quad \phi_1 + \phi_2 = 1,$

where $\frac{\partial \hat{f}}{\partial t} = -\frac{\lambda_f}{\rho_0} \tilde{\omega}_2 \nabla \cdot \vec{J}_2^D + \lambda_f \varepsilon \nabla \cdot \left(\nabla \phi_2 \frac{\partial \phi_2}{\partial t} \right),$

$$\tilde{\omega}_2 = \omega_2 - \omega_1.$$

Next, we use the identity

$$-\tilde{\omega}_2 \nabla \cdot \vec{J}_2^D = -\nabla \cdot (\tilde{\omega}_2 \vec{J}_2^D) + \vec{J}_2^D \cdot \nabla \tilde{\omega}_2,$$

to obtain

$$\begin{aligned}
\frac{\partial \hat{f}}{\partial t} &= \frac{\lambda_f}{\rho_0} \nabla \tilde{\omega}_2 \cdot \vec{J}_2^D - \frac{\lambda_f}{\rho_0} \nabla \cdot (\tilde{\omega}_2 \vec{J}_2^D) \\
&\quad + \lambda_f \varepsilon \nabla \cdot \left(\nabla \phi_2 \frac{\partial \phi_2}{\partial t} \right) \\
&= -\nabla \cdot \vec{J}_f^D + \sigma_f
\end{aligned}$$

where

$$\vec{J}_f^D = \frac{\lambda_f}{\rho_0} \tilde{\omega}_2 \vec{J}_2^D - \lambda_f \varepsilon \nabla \phi_2 \frac{\partial \phi_2}{\partial t}$$

$$\sigma_f = \frac{\lambda_f}{\rho_0} \nabla \tilde{\omega}_2 \cdot \vec{J}_2^D$$

We will take

$$J_2^D = -M \nabla \tilde{\omega}_2$$

so that

$$\sigma_f = -\frac{\lambda_f}{\rho} M |\nabla \tilde{\omega}_2|^2.$$

Thus,

$$\rho_0 \frac{\partial \phi_2}{\partial t} = M \Delta \tilde{\omega}_2,$$

where

$$\tilde{\omega}_2 = \frac{1}{\varepsilon} \tilde{W}'(\phi_2) - \varepsilon \Delta \phi_2,$$

and

$$\tilde{W}(\phi) = W(1-\phi, \phi).$$

This is the isothermal Cahn-Hilliard equations.